

TrES-3b

R-1929

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140506 · created 2026-08-06T01:32:26+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456783.9169 +/- 0.00099 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1966 +/- 0.0055
Transit depth (Rp/Rs)^2	3.87 +/- 0.22 [%]
Orbital Inclination (inc)	83.39 +/- 0.58
Airmass coefficient 1 (a1)	1.558 +/- 0.017
Airmass coefficient 2 (a2)	-0.424 +/- 0.062
Scatter in the residuals of the lightcurve fit is	1.08 %
Best Comparison Star	#2 - [27, 310]
Optimal Aperture	4.0
Optimal Annulus	8.67
Transit Duration (day)	0.0698 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1966 $\pm$ 0.0055 vs 0.16309 (deviation 5.53 σ)
Duration fit vs published	0.0698 d vs 0.0567 d (deviation 3.31 σ)
Mid-time O-C	1.1 min
Depth SNR	32.5
Residual scatter	1.08 %

Light curve

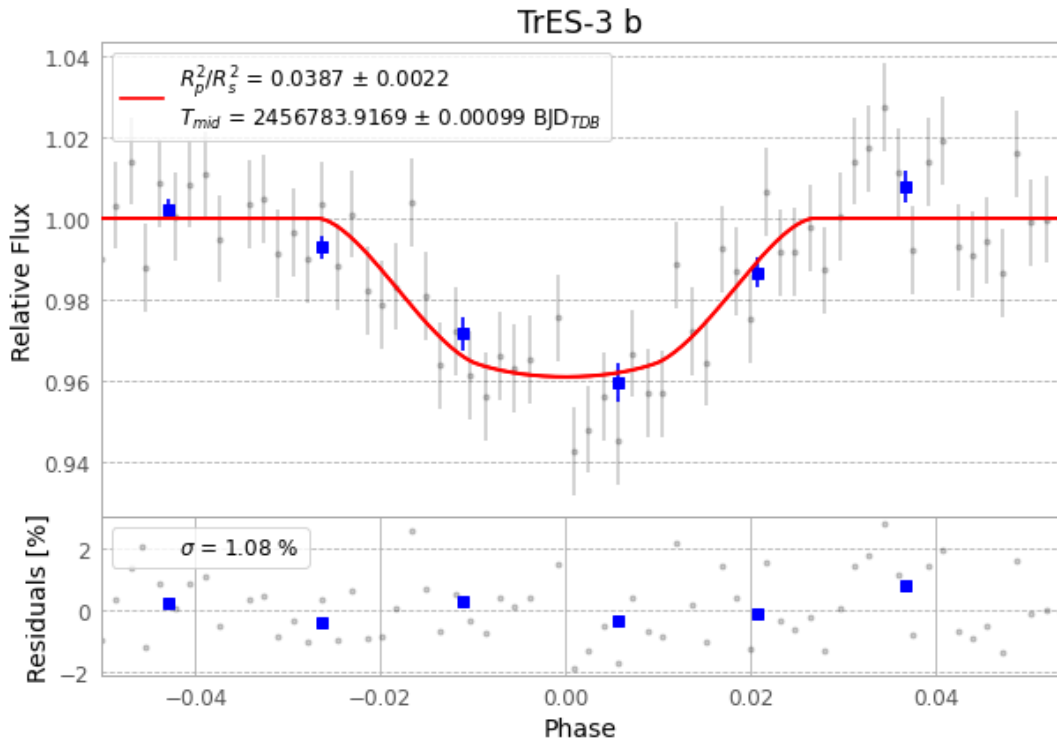


Figure 1 — FinalLightCurve_TrES-3 b_2014-05-06.png (SHA-256 1350877a2609224d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 265], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0db55abf10dbfcb9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-3140506082112.FITS → TRES-3140506113612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140506.log (3bd3ed342e98...), FinalLightCurve_TrES-3 b_2014-05-06.png (1350877a2609...), FinalLightCurve_TrES-3 b_2014-05-06.csv (128ff07f2a4c...)

Compute resources

Wall time 194.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 01:32Z.